

WINDOWS PROCESS CREATION INVESTIGATION

POWERSHELL ACTIVITY ANALYSIS

SOC Analyst Home Lab Project

Prepared By:
Benard Kiprotich

Date:
June 2026

Project Overview

This project demonstrates the investigation of Windows **Process Creation Events (Event ID 4688)** within a controlled cybersecurity home lab. The objective was to monitor process execution, analyze parent-child process relationships and determine whether the observed activity represented legitimate system administration or potential malicious behavior.

During the investigation, Windows PowerShell was used to execute several native Windows utilities, including `whoami.exe`, `hostname.exe`, `ipconfig.exe`, and `cmd.exe`. Each process creation event was examined using Windows Event Viewer to reconstruct the execution timeline and understand how the processes were related.

The collected evidence was analyzed using standard Security Operations Center (SOC) investigation techniques. Parent-child process relationships were correlated, observed behaviors were mapped to the MITRE ATT&CK framework and the activity was assessed based on its context within the laboratory environment.

The investigation concluded that the observed process executions were **benign**, as they were initiated intentionally by an authorized user during a controlled cybersecurity exercise. Although the commands are commonly used by attackers during reconnaissance, the surrounding evidence indicated no signs of unauthorized access or malicious activity.

Process Creation Investigation: PowerShell Activity Analysis

Executive Summary

During routine monitoring of Windows Security Event Logs, multiple **Event ID 4688 (Process Creation)** events were identified involving PowerShell launching several Windows system utilities.

The objective of this investigation was to analyze the process execution chain, determine whether the observed activity was malicious or benign, and document the findings using standard Security Operations Center (SOC) investigation procedures.

Analysis determined that the activity was **benign** and consistent with authorized user actions performed in a controlled cybersecurity home lab.

Incident Information

Field	Value
Investigation ID	SOC-2026-002
Investigation Type	Process Creation Analysis
Event ID	4688
Severity	Informational
Status	Closed
Analyst	Benard Kiprotich
Date	June 2026

Objective

The objectives of this investigation were to:

- Monitor Windows Process Creation Events.
- Investigate parent-child process relationships.
- Identify potentially suspicious PowerShell activity.
- Correlate multiple Event ID 4688 records.
- Document findings using a professional SOC investigation methodology.

Lab Environment

Component	Details
Operating System	Windows 10 Virtual Machine
Hypervisor	Oracle VirtualBox
Log Source	Windows Security Log
Event Viewer	Windows Event Viewer
Event Investigated	4688
Supporting System	Kali Linux

Investigation Timeline

Step	Activity
1	User opened Windows PowerShell
2	PowerShell executed whoami.exe
3	PowerShell executed hostname.exe
4	PowerShell executed ipconfig.exe
5	PowerShell launched cmd.exe
6	Command Prompt executed dir
7	User exited Command Prompt
8	User exited PowerShell

Evidence Collected

Event 1

Field	Value
Event ID	4688
Process	whoami.exe
Parent Process	powershell.exe
Purpose	Identify current logged-on user

Event 2

Field	Value
Event ID	4688
Process	hostname.exe
Parent Process	powershell.exe
Purpose	Retrieve computer hostname

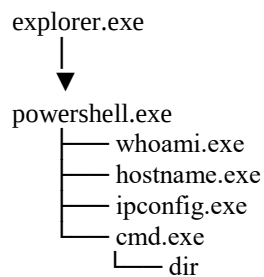
Event 3

Field	Value
Event ID	4688
Process	ipconfig.exe
Parent Process	powershell.exe
Purpose	Display network configuration

Event 4

Field	Value
Event ID	4688
Process	cmd.exe
Parent Process	powershell.exe
Purpose	Launch Windows Command Prompt

Process Tree



Analysis

The investigation revealed that PowerShell served as the parent process for several command-line utilities. The executed commands collected basic information about the logged-on user, system name, and network configuration before launching the Windows Command Prompt.

These commands are commonly used by:

- System administrators
- IT support personnel
- Cybersecurity analysts
- Threat actors performing reconnaissance

While the commands themselves are not malicious, they are frequently observed during the early stages of an intrusion. Therefore, analysts should evaluate them alongside other evidence such as user context, command-line arguments, network activity, and subsequent process execution.

In this investigation, all activity was initiated by an authorized user within a controlled home lab environment. No evidence of unauthorized access, persistence mechanisms, or malicious payload execution was identified.

MITRE ATT&CK Mapping

Technique ID	Technique	Evidence
T1059.001	PowerShell	powershell.exe
T1059.003	Windows Command Shell	cmd.exe
T1033	System Owner/User Discovery	whoami.exe
T1082	System Information Discovery	hostname.exe
T1016	System Network Configuration Discovery	ipconfig.exe

Indicators of Compromise (IOCs)

No indicators of compromise were identified during this investigation.

Observed processes were legitimate Windows utilities executed by an authorized user within a controlled laboratory environment.

Detection Opportunities

Security teams can improve visibility into similar activity by:

- Monitoring Event ID 4688 for PowerShell process creation.
- Alerting on unusual parent-child process relationships.
- Enabling command-line process logging for richer forensic analysis.
- Correlating PowerShell execution with network connections and file creation events.
- Investigating PowerShell processes spawned by applications such as Microsoft Office, email clients, or web browsers.

Recommendations

1. Enable command-line process auditing to capture process arguments.
2. Forward Windows Security logs to a SIEM for centralized monitoring.
3. Establish a baseline of normal administrative PowerShell usage.
4. Investigate unexpected PowerShell execution chains, especially when initiated by user-facing applications.
5. Continue monitoring process creation events to detect abnormal execution patterns.

Conclusion

This investigation demonstrated how Windows Security Event ID 4688 can be used to analyze process creation and parent-child process relationships. Multiple PowerShell-launched processes were successfully identified and correlated into a single execution timeline.

The observed activity was determined to be **benign** and consistent with authorized user actions performed in a home lab. The project demonstrates foundational SOC analyst skills in Windows event log analysis, process investigation, threat assessment, MITRE ATT&CK mapping, and incident documentation.